Chapter 5, Section 6

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1. Let X be a surface in $\mathbb{P}^n$, $n \geq 3$, defined as the complete intersection of hypersurfaces of degree $d_1, \ldots, d_{n-2}$, with each $d_i \geq 2$. Show that for all but finitely many choices of $(n, d_1, \ldots, d_{n-2})$, the surface X is of general type. List the exceptional cases, and where they fit into the classification picture.

Proof. We will use (II, Ex. 8.4e), which states the canonical sheaf ω_X is isomorphic to $\mathcal{O}_X(\sum_{i=1}^{n-2} d_i - n - 1)$, and assume d_i is in descending order. Since $d_i \geq 2$, for every $(n, d_1, \ldots, d_{n-2})$ with $n \geq 6$, the inequality

$$\sum_{i=1}^{n-2} d_i - n - 1 > 0$$

always holds, which implies ω_X is very ample, hence X is of general type. If $n = 3$ and $d_1 \geq 5$, then X is of general type. If $n = 4$ and either $d_1 \geq 4$ or $d_2 \geq 3$, then X is of general type. If $n = 5$ and $d_1 \geq 3$, then X is of general type. Thus, the exceptional cases are:

$\boxed{n = 3, d_1 = 2}$ X is the nonsingular quadric surface, isomorphic to $\mathbb{P}^1 \times \mathbb{P}^1$, and it is also the rational ruled surface with invariant $e = 0$.

$\boxed{n = 3, d_1 = 3}$ X is a cubic surface, which is isomorphic to $\mathbb{P}^2$ with 6 points in general position blown up, and it is also called a Del Pezzo surface of degree 3.

$\boxed{n = 3, d_1 = 4}$ In this case, $\omega_X \cong \mathcal{O}_X$, and we have a short exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^3}(-4) \longrightarrow \mathcal{O}_{\mathbb{P}^3} \longrightarrow \mathcal{O}_X \longrightarrow 0 \ ,$$

and the long exact sequence in cohomology gives

$$\mathrm{H}^1(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}) \longrightarrow \mathrm{H}^1(X, \mathcal{O}_X) \longrightarrow \mathrm{H}^2(\mathbb{P}^3, \mathcal{O}_{\mathbb{P}^3}(-4)) \ ,$$

so $q = h^1(X, \mathcal{O}_X) = 0$. Hence, X is a K3 surface.

$\boxed{n = 4, d_1 = d_2 = 2}$ X is a Del Pezzo surface of degree 4 with $\omega_X \cong \mathcal{O}_X(-1)$, which is isomorphic to $\mathbb{P}^2$ with 5 points blown up (Ex. 4.13).

$\boxed{n = 4, d_1 = 3, d_2 = 2}$ In this case, $\omega_X \cong \mathcal{O}_X$. Let $\mathcal{I}$ be the ideal sheaf of X in $\mathbb{P}^4$. The long exact sequence in cohomology gives the isomorphism

$$\mathrm{H}^1(X, \mathcal{O}_X) \cong \mathrm{H}^2(\mathbb{P}^4, \mathcal{I}).$$

By (III, §7), we can use the Koszul resolution of $\mathcal{I}$ to compute its cohomology

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^4}(-5) \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-3) \oplus \mathcal{O}_{\mathbb{P}^4}(-2) \longrightarrow \mathcal{I} \longrightarrow 0 \ ,$$

which readily shows $\mathrm{H}^2(\mathbb{P}^4, \mathcal{I}) = 0$. Hence, X is a K3 surface.

$\boxed{n = 5, d_1 = d_2 = d_3 = 2}$ In this case, $\omega_X \cong \mathcal{O}_X$. Let $\mathcal{I}$ be the ideal sheaf of X in $\mathbb{P}^5$. The long exact sequence in cohomology gives the isomorphism

$$\mathrm{H}^1(X, \mathcal{O}_X) \cong \mathrm{H}^2(\mathbb{P}^5, \mathcal{I}).$$

By (III, §7), we can use the Koszul resolution of $\mathcal{I}$ to compute its cohomology

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-6) \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-4)^{\oplus 3} \longrightarrow \mathcal{O}_{\mathbb{P}^5}(-2)^{\oplus 3} \longrightarrow \mathcal{I} \longrightarrow 0 \ ,$$

which readily shows $\mathrm{H}^2(\mathbb{P}^5, \mathcal{I}) = 0$. Hence, X is a K3 surface. □

2. Prove the following theorem of Chern and Griffiths. Let X be a nonsingular surface of degree d in $\mathbb{P}^{n+1}_{\mathbb{C}}$, which is not contained in any hyperplane. If $d < 2n$, then $p_g(X) = 0$. If $d = 2n$, then either $p_g(X) = 0$, or $p_g(X) = 1$ and X is a K3 surface.

Proof. □